

How Biology can Control Information Flow in the Nervous System

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Abstract

In this note, the author proposes a definition for neurobiological consciousness. The definition is based on a thought experiment wherein the activity of neuronal ion pumps is intervened with, leading to a modulation in information processing. The view is thus a bottom-up view where the physical layer of the system influences the data layer.

The transmembrane potential difference between the intracellular medium and the extracellular medium around a neuron is maintained by ion channels and ion pumps. The ion pump can be modulated in its activity by removing its energy source or by the use of an inhibitor [1]. With this modulation we acquire control over the potential difference that is maintained in steady state which in turn is a reflection of the maintained internal and external ion concentrations.

Since the ions in the extracellular space conduct electricity, alterations in their concentrations ought to impact the conductivity of the extracellular medium. Thus, by use of an inhibitor we can impact the extracellular conductivity. Now the ephaptic interaction between axons strengthens when there is a rise in resistivity of the extracellular space [2]. When this interaction is strong, synchronization takes place more rapidly than otherwise. Since ephaptic synchronization between axons leads to an amplification of signals over noise [3], the sensitivity of the nervous system to impinging information can be modulated by the use of an inhibitor. Thus, ion pump biology can control the flow and processing of information in the nervous system.

Inspired by this interventionist view, neurobiological consciousness may be defined as the ability to up- and down- regulate the activity of the ion pumps in the nervous system. Organisms which are able to thus self-regulate, may be said to be conscious, with greater control being a result of a greater manifestation of consciousness.

References

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